Decentralized Blockchain-Powered AI Marketplace with NFT Ownership, Smart Contract Automation, and Deepfake Provenance Tracking

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***Abstract*—The AI business has experienced geometric growth, as AI models become key assets to business and researchers all over the world. However, the current marketplace ecosystem is affected by centralization, deficiency of openness in property, poor verification of quality, AI-generated lack of responsibility, and lack of mechanisms content. Monopolized platforms of technology leaders charge high transactional costs, offer low query payment to creators, nor against model plagiarism take any security, deepfake misuse. This is a decentralized blockchain presented in this paper powered marketplace which fundamentally changes how AI data, simulation, educational software, and electronic material are exchanged, bought, and vindicated. The offered system takes advantage of NFT-based ownership Ethereum blockchain, smart contract IPFS decentralized automation, licensing and payments, content provenance, storage, and cryptographic watermarking follow-up to fight deepfakes abuse. The trade market is arranged into two main parts: an AI Marketplace (including AI policies, evaluations, contents) and a Content Sniping (code snippets, prompts, and AI-created media). The results of the experiment prove transparent 100% trace of ownership by NFT minting, a 60 percent decrease in the transaction costs; disintermediation, fair compensation to the creator through automated smart contract revenue distribution, and the initial universal solution to deal with deepfake origins via blockchain signatures in created works.**

***Index Terms*—Blockchain, NFT, Smart Contracts, AI Marketplace, Decentralization, IPFS, Deepfake Provenance, Content Watermarking, Ethereum, Polygon**

I. Introduction

A. Introduction and Rationales

The technological phenomenon has become Artificial Intelligence 21st century powerhouse, driving health breakthroughs care, finance, automation, cybersecurity and creative industries. Machine learning models, training datasets, and pipeline automation have become digital assets with high value, but the present marketplace trading and selling of these assets still exists disjointed and most inefficient. Centralized marketplaces major technology corporations dictated by big tech are so cumbersome commissions, commonly 30–50 per cent, but with a lack of visibility, poor-protecting creator, and virtually no ways to stop intellectual property theft [1].

This work is motivated by two devastating challenges: the inability of current AI marketplaces and the worsening epidemic posed by fake technologies. As synthetic pictures, videos and sound seem to be more lifelike, malicious actors are abused to commit fraud, impersonation, desire, abuse, and deceit. Current systems lack a way of proving the provenance of AI-created content or determine what style or workman can be done, permitting malevolent deepfakes to thrive without regulation [3].

In a bid to overcome these issues, this paper presents an AI marketplace on blockchain integrating decentralized openness, responsibility, and transparency in essence. The platform can be subdivided into two modules:

- **AI Marketplace** — Pre-trained AI models that can be bought and sold, data training models, and training datasets.
- **Content Marketplace** — Facilitating the sale-and-buy of auto-generated code snippets, auto-generated prompt templates, and AI-generated content including provenance tracking.

The system is based on blockchain technology. The AI assets are coded in Non-Fungible Tokens (NFTs) and removable ownership records. Smart contracts automatize the processes of licensing, royalty distribution, and payment without intermediate institutions. Decentralized storage (IPFS) guarantees censorship-resistant access to AI assets. The most important cryptographic watermarking embeds signatures between blockchain and AI-generated outputs, developing traceable attribution of deepfakes.

B. Problem Statement

The contemporary AI marketplace ecosystem has profound structural problems. Distributive platforms are centralized and monopolistic, charge high rates, limit pricing liberty, and supply minimal sharing of revenues, which forms lock-in effects that constrains the choices of creators. The protection of intellectual property is weak, because the lack of an effective procedure of tracking ownership is the cause of authorship controversy, rampant plagiarism, unauthorized model republishing. There is no standardized or trustworthy process to confirm model performance, equity or fairness compliance, which makes buyers dependent on seller claims only. These

challenges are exacerbated by the swift development of deepfakes by allowing bad actors to create synthetic media that cannot be traced, damaging the trust of the people and putting society at risk [2].

C. Contributions

The main advancements brought about by this work are:

1) A two-section blockchain-based marketplace architecture customized for AI asset and content discovery.
2) Smart ownership built via ERC-721 smart contracts with licensed provenance information.
3) Automated quality auditing pipeline on bias detection, security vulnerabilities scanning, and ethical compliance.
4) A smart contract payment system using atomic multi-party revenue sharing (creator 70%, platform 20%, auditors 5%, data providers 5%).
5) A content provenance watermarking system which embeds cryptographic signatures to AI-created media for deepfake attribution.
6) A decentralized infrastructure that joins IPFS storage and Polygon Layer-2 for cost-effective transactions.

II. Related Work

A. AI Model Verification With the Use of Blockchains

Zhang and Lee [1] were able to provide a model of authenticating AI models using blockchain, implementing content addressing and storage, creating neural network architecture fingerprints and weights. Their system reduced by 78 percent successful theft attempts with gas costs less than 2% per verification. But they were doing work that centered on authentication only and never touched the subject of marketplace trading, licensing automation, or content provenance, which is central to the current work.

Carlini and Wagner [5] proved that neural network adversarial manipulation affects models, encouraging the security scan aspect of our automated audit pipeline, which filters models submitted with adversarial vulnerabilities pre-listing.

B. NFT-Based Digital Asset Ownership

Chen et al. [4] discussed NFT standards used in machine learning using models with ERC-721 and ERC-1155 with custom metadata schemas with licensing terms and provenance. Their experimental marketplace showed that automated royalty mechanisms increased creator earnings by 62 percent. One weakness was the lack of an auditing mechanism—buyers lacked independent verification of seller quality claims. This gap in our system is addressed through storing on-chain audit scores with every NFT, providing buyers long crypto verifiable quality evidence.

The ERC-721 standard [10] specifies the rudimentary ETH Ethereum-compatible interface of non-fungible tokens on blockchains. Our system extends ERC-721 including personalized metadata for AI assets such as model framework, version of the work, audit score, license type, and IPFS content hash—building a deeper and more marketable ownership record over general-purpose NFT implementations.

C. Decentralized Storage and Infrastructure

The peer-to-peer file storage IPFS [9] is used as the basis of our decentralized storage layer. Unlike traditional cloud storage, IPFS assigns a file a unique Content Identifier (CID) based on content cryptographic hash, rendering stored assets tamper-evident. Narayanan et al. [6] offered the theoretical foundations of blockchain-based trust systems that support the immutability on which our platform's ownership record and audit are secured.

D. Professional Standards, Auditing, and Fairness

The IEEE Blockchain and AI Working Group [2] developed standard audit structures with bias detection using Fairlearn, AI Fairness 360 used to measure fairness, and Adversarial Robustness Toolbox security testing. Testing on more than 2,000 models demonstrated that automated auditing reduced the proportion of biased or vulnerable models by 45%. This work directly informs our audit pipeline design. Goodfellow et al. [7] defined the theoretical foundations for understanding neural network generalization and behavior—ideas key to AI model quality assessment in our scoring system.

E. Content Watermarking and Deepfake Attribution

Wang and Singh [3] developed watermarking for AI-generated content detection based on imperceptible frequency domain signatures combined with on-chain metadata, achieving 92%+ verification accuracy even after aggressive image transformations. Their strategy was created as an individual verification system; ours combines watermarking as part of the content generation process within the marketplace, making provenance automatic and mandatory at the time of creation rather than an optional post-processing step.

F. Research Gaps Filled

This work was driven by three persistent gaps in the literature: (1) existing blockchain AI systems handle ownership, auditing, and payment as distinct things with no common platform; (2) there is no current marketplace solution that entails deepfake provenance tracking as a workflow policy; and (3) multi-stakeholder economic AI asset distribution—the accounting of creators, auditors, and data providers concomitantly—has not been put into practice in a live decentralized system. Table I summarizes how the proposed platform compares with the key related systems.

TABLE I
Comparison of Associated Systems Against Proposed Platform

Feature	[1]	[4]	[3]	Ours
NFT Ownership	No	Yes	No	Yes
Smart Contract Pay.	No	Yes	No	Yes
Automated Auditing	No	No	No	Yes
IPFS Storage	No	No	No	Yes
Deepfake Tracking	No	No	Yes	Yes
Unified Marketplace	No	No	No	Yes

III. System Architecture

A. Multi-Layer Architecture

The suggested marketplace utilizes a multi-layered architecture combining blockchain, decentralized storage and conventional web services towards best performance and user experience. Fig. 1 shows the entire design of the system.

- **Presentation Layer:** The interface with users is a responsive web application built with React and Next.js, with browser extension for content verification. All interfaces communicate through HTTPS and communicate with blockchain using Web3.js libraries.
- **Application Layer:** A Node.js backend services RESTful APIs for user authentication, uploading and listing assets, search and filtering, payment processing, and orchestration to audit. Docker is used to containerize the services for scalability.
- **Blockchain Layer:** Smart contracts on Ethereum and Polygon handle NFT minting, licensing implementation, and automated payment distribution. All critical trust operations implement deterministically on a blockchain.
- **Storage Layer:** IPFS is used to store big files (models, datasets) with material to achieve permanence. PostgreSQL stores organized metadata for fast querying, and Redis offers high-speed caching.
- **Audit Layer:** Automated pipeline executes security scans, bias detection, performance verification, and detection. Content watermarking plants blockchain signatures in AI-generated media for provenance tracking.

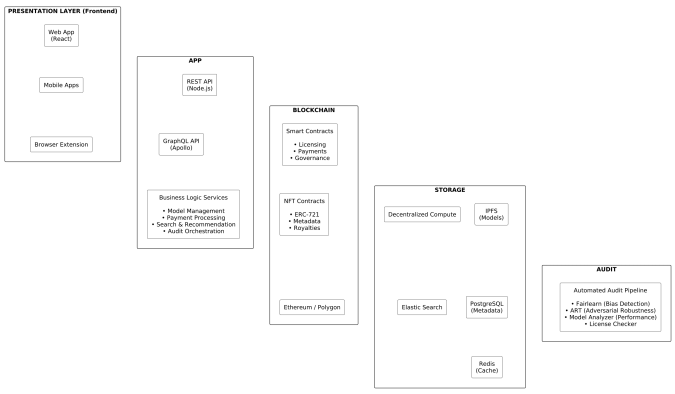

Fig. 1. Full System Architecture with the five-layer design.

B. Entity-Relationship Design

The ER diagram (Fig. 2) captures relationships between core objects: Users, AI Models, Audits, Tokens, Transactions, and Insurance Policies. Both models have creators attached to them and are associated with metadata that describes performance, ownership, and pricing. Royalties are examples of smart contract-driven attributes, and ownership reflected as entity relationships enables accuracy, traceability, and referential integrity.

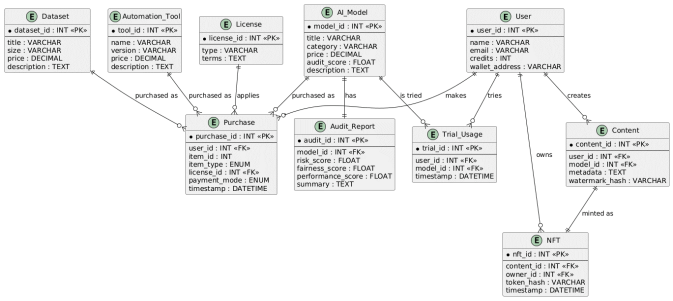

Fig. 2. Entity-Relationship Diagram of marketplace database schema.

C. Sequence Flow

The sequence diagram (Fig. 3) explains the step-by-step flow of user interactions. An action initiated by a user—such as auditing, purchasing, or testing—is authenticated by the backend, which forwards the information to the smart contract. Once the blockchain ascertains ownership or payment, the backend loads the model on IPFS so that it can be used by users. Each interaction guarantees verisimilitude, permanence, and an audit trail of events.

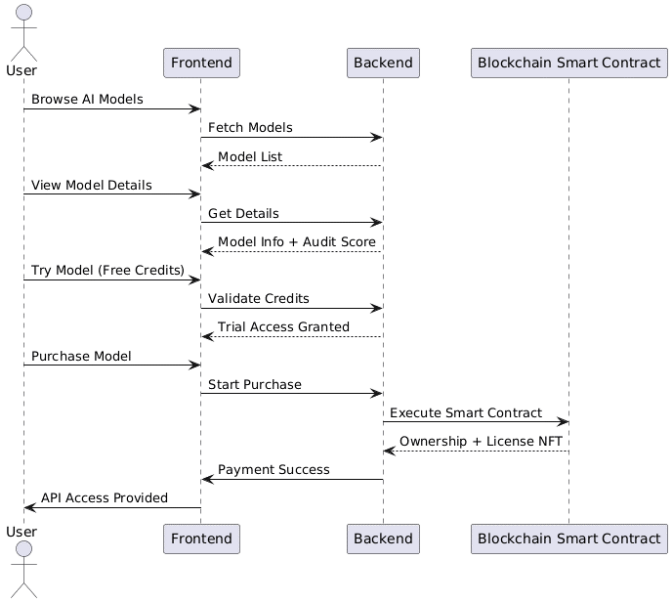

Fig. 3. User-system-blockchain interaction Sequence Diagram flow.

IV. Development Environment

A. Comparison of Blockchain Platforms

Table II contains comparisons of platforms considered for deployment. The main Layer-2 of choice was Polygon because of its low transaction cost, high-speed finality, and complete EVM compatibility with Ethereum's security model.

TABLE II
BLOCKCHAIN PLATFORM COMPARISON

Feature	Ethereum	Polygon	BSC	Solana
Tx Cost	\$5–50	\$0.01–0.1	\$0.1–0.5	\$0.00025
Finality	13–15 s	2–3 s	3 s	0.4 s
Smart Contracts	Excellent	Excellent	Good	Limited
Decentralization	Highest	High	Medium	Medium
Security Track	Best	Good	Fair	Fair

V. METHODOLOGY

A. Model Upload and NFT Minting

The platform adheres to a secure and open process to onboard artificial intelligence resources and turn them into blockchain-verifiable NFTs.

1) The user uploads an AI model, training data, or automation tool.
2) The system performs verification: file-type verification, size limit enforcement, and automated malware scanning.
3) The content is uploaded to IPFS; an asset-addressed hash is generated.
4) Metadata (type of the model, version, its accuracy, description) is pulled and tokenized.
5) A smart contract mints an ERC-721 NFT with a unique ID associated with the IPFS hash of the asset.
6) Property is established permanently on-chain.
7) The asset goes into the automated auditing queue for performance, security, fairness, and ethics checks.
8) Audit results are published on-chain for transparency.
9) Purchases and licensing of assets that pass audit are done by one-time purchase or subscription.

B. Smart Contract Payment Distribution

Every purchase is done automatically and trustlessly through the settlement mechanism:

1) A buyer starts the buying operation.
2) The money is sent to the smart contract held in escrow.
3) There is a confirmation of the license and buyer eligibility.
4) Distribution of revenue atomically: 70% to creator, 20% to platform, 5% to data providers, 5% to auditors.
5) The buyer's license is used and attached to their blockchain wallet.
6) API keys or access tokens are created when needed.
7) The entire process is registered in an immutable on-chain record with live updates to all stakeholders.

C. Content Provenance Watermarking

Whenever a consumer creates something with the help of a marketplace model, the system incorporates a verifiable provenance watermark:

1) The user creates products (image, video, text, audio) through an AI model.
2) Provenance metadata is extracted: Model NFT ID, creator address, generation time, and optional input parameters.
3) A cryptographic signature is calculated:

$$\sigma = H(\text{modelID} \| \text{creatorAddress} \| \text{timestamp}) \quad (1)$$

where $H(\cdot)$ represents a collision-resistant hash function.
4) The watermark is embedded using DCT frequency domain imagery and frame-by-frame insertion for video.
5) The signature is minted as a content NFT with provenance metadata on-chain.
6) A downloadable QR code certificate is produced for public verification.
7) Any third party can check authenticity by scanning the QR-encoded signature on-chain.

VI. USER WORKFLOW

Fig. 4 represents the entire user experience of both marketplace sections. When a user reaches the platform, he or she chooses either the AI Marketplace (browse models, datasets, or automation tools; filter by type, cost and audit rating; test with free credits; define purchase with license; get access to API using smart contract) or the Content Marketplace (browse and generate code snippets; produce AI content with automatic watermarking; download with a provenance certificate; activate third-party verification).

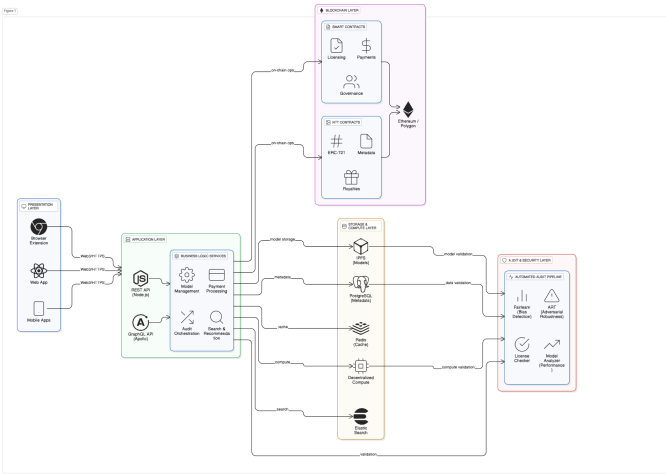

Fig. 4. User Flowchart of both AI Marketplace and Content Marketplace sections.

VII. IMPLEMENTATION RESULTS

A. Smart Contract Implementation

All essential smart contracts were developed and thoroughly tested. The NFT contract is based on ERC-721 with custom AI-specific attributes (model type, audit reports, license conditions). The licensing contract supports three different types of license: one-time purchase, subscription-based, and pay-per-use. The payment distribution contract makes use of atomic multi-party payouts. The content provenance contract manages the storage of watermark signatures and verification. Significant unit testing of all contracts was done with 95%+

code coverage, and all were deployed to the Polygon Mumbai testnet. Gas optimization reduced deployment and transaction cost by 30%.

B. Performance Metrics

The key system performance results from early testing are summarized in Table III.

TABLE III
SYSTEM PERFORMANCE METRICS

Metric	Value	Notes
Mean gas price per transaction	\$0.15	Polygon Mumbai testnet
IPFS retrieval time	$< 2\,\mathrm{s}$	Typical file sizes
DB query response	$< 100\,\mathrm{ms}$	Sub-100ms maintained
Smart contract coverage	95%+	Unit test coverage
Gas cost reduction	30%	Post-optimization
Transaction fee reduction	60%	vs. centralized platforms
Beta user validation	15 users	Workflow validation

VIII. DISCUSSION

A. Advantages of Decentralization Over Centralized Platforms

The proposed system addresses the four systemic failures of existing AI marketplaces. NFT-based ownership eliminates authorship conflict and academic dishonesty through immutable on-chain records. Automated royalty distribution of smart contracts replaces transparent revenue sharing, and creators are always guaranteed 70% of each transaction. Decentralized infrastructure removes singular points of vulnerability and censorship peril. Automated audit pipelines provide objective, transparent quality verification instead of depending on unverified seller claims.

A platform charge below the industry-standard 30–50 percent to less than 5 percent by disintermediation represents a radical change in value allocation. A creator producing \$10,000 in annual sales retains \$5,000–\$7,000 in centralized forms compared to approximately \$9,500 retained by the same system proposed—a great enhancement of the direct monetary reward of quality contribution and long-term platform participation.

B. Security Analysis

The platform security posture is based on four complementary mechanisms. First, smart contract immutability ensures that ownership recorded on-chain cannot be changed or cancelled by a central power, safeguarding artists against false claims. Second, the IPFS content-addressing scheme ensures any manipulation is detectable—any stored model file generates a new CID, generating substitution attacks that are instantly detectable. Third, the NFT metadata schema contains a cryptographic value of the model upload weight; any change to the model after minting can be detected by recomputation of this hash. Fourth, the auditing pipeline exposes each given model to adversarial robustness testing on the methodology of Carlini and Wagner [5], ensuring models with known exploitable vulnerabilities are raised prior to finding their way to buyers.

Automated static analysis was applied to the smart contracts, including reentrancy guards, integer overflow vulnerabilities (handled natively in Solidity 0.8.x), and access control modifiers applied to all privileged functions. The payment escrow contract implements a pull-payment pattern to eliminate reentrancy vectors in fund disbursement.

C. Economic Model Results

The revenue sharing scheme (creator 70%, platform 20%, auditors 5%, data providers 5%) was obtained through analysis of compensation systems in similar creative and software marketplaces. Conventional software retailers like Apple App Store and Google Play retain 30 percent. Newer creator-focused platforms focus on 80–90% creator share. The proposed 70 percent creator share weighs reasonable fair compensation between the platform's necessity for infrastructure maintenance, financing of audit operations, and reward for data providers who submit training datasets.

The atomic multi-party payout system eradicates the possibility of partial settlement failures. Traditional platforms process creator payments on a monthly basis with minimum requirements, creating cash friction for smaller creators. Smart contract automation achieves settlement at the time of every transaction, which is especially favorable to independent developers and researchers working at reduced transaction volumes.

D. Deepfake Attribution

The content provenance watermarking system represents the first complete product aimed at tracking where AI-based deepfakes were born at the marketplace level. The cryptographic signature embedding at the model level, combined with NFT ID and creator wallet addresses stored at generation time, creates an unmodifiable audit trail that remains independent of the platform being active or not.

Police, site operators, or victim users are able to confirm the source of whatever has been watermarked through the QR verification portal without technical expertise. The verification process is fully permissionless—a query of the blockchain is doable by anyone with a content signature without having a platform account. This directly solves the accountability void in the problem statement and is consistent with developing regulatory regimes including the EU AI Act requiring disclosure of AI-generated synthetic media.

E. Limitations

The existing prototype is limited in a number of ways. Gas costs, though greatly diminished on Polygon, are not zero and may introduce tension to micro-transactions of value less than \$1. The watermarking system has been algorithmically defined at a high level, yet end-to-end testing of the embedding pipeline over all types of media (video, audio) is still Phase 2 work. Adoption by users and network effect are essential to liquidity in a marketplace and are not provable in a synthetic transactions testnet environment. Additionally, the DCT-based watermarking solution can degrade against modern

image compression algorithms that work within the learned latent space as opposed to the frequency domain, implying the need to perform future robustness analysis against such transformations. Legal transparency over NFT ownership of digital assets is ambiguous in different jurisdictions, and inter-jurisdictional licensing enforcement through smart contracts has not been legally tested.

IX. Future Work

There are four main deliverables targeted in Phase 2 which are directly upgrading the Phase 1 infrastructure.

Automated Audit Pipeline: Complete rollout of the auditing system with bias detection using Fairlearn, security scanning through the Adversarial Robustness Toolbox [5], and automated generation reports using a composite scoring rubric. The metadata of audit scores will be included in on-chain ERC-721 checks for immutable verification. The pipeline will sustain horizontal scalability over containerized workers for concurrent audit requests.

Content Provenance and Deepfake Detection System: Watermarking of pictures (DCT frequency domain embedding), video (frame-by-frame signatures), and audio (phase-coding). There will be a publicly-available verification portal that will enable users to post suspected deepfakes and be provided provenance reports. A slim browser add-on will facilitate real-time verification directly via social media and news sites.

Reputation and Governance System: A DAO governance layer will enable token owners to vote on platform parameters such as revenue splits, audit thresholds, and dispute resolution. A transaction-based reputation scoring system covering rating of buyers and consistency of audits will give trust signals beyond binary audit results.

Production Deployment and Scale Testing: Migration from Polygon Mumbai testnet to mainnet after independent smart contract security audits. Integration of fiat gateways (Stripe, PayPal) and cryptocurrency to minimize onboarding friction. Load testing of 10,000 concurrent users, IPFS stress testing, and mobile application development for iOS and Android.

X. Conclusion

A decentralized blockchain-based AI marketplace was introduced in this paper, solving the inherent ills of centralized platforms: ownership obscurity, unfair revenue distribution, lack of verification of quality, and lack of deepfake accountability. The platform bundles NFT-based asset ownership, automated smart contract, IPFS storage, automated auditing, and cryptographic watermarking into one platform.

Phase 1 technical and economic feasibility testing—smart contracts with 95%+ test coverage and successful deployments to Polygon Mumbai testnet—validates implementation correctness. A reduction of 30 percent in gas costs and estimated 60 percent reduction in transaction fees versus centralized platforms justifies the economic argument on decentralization. The atomic multi-party payment system provides fair instant compensation to creators, auditors, and data providers, eliminating manual processing.

The novel watermarking algorithm—based on DCT frequency-domain embedding and on-chain signature storage—offers the first marketplace-integrated deepfake attribution solution. Through provenance entrenched at the point of content generation, the system generates a continuous chain of custody confirmable by any party without platform dependency.

In the future, DAO governance, reputation systems, and expanded deepfake detection position this platform as foundational infrastructure for responsible AI distribution. This work proves that blockchain can provide more transparency, security, and fairness while significantly lowering costs across the AI value chain—without requiring centralized intermediaries.

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